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DEPARTMENT OF STATISTICS

M.Sc. QUANTITATIVE FINANCE
MSQF 532: FINANCIAL MATHEMATICS

REPORT ON
EQUITY VALUATION
INDIAN FMCG COMPANIES

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SEPTEMBER 2026

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1 Key Analytical Assumptions, Data Limitations, and Valuation Framework

1.1 Methodological Assumptions and Data Reconciliation

The cross-sectional empirical valuation of ten premier Indian Fast-Moving Consumer Goods (FMCG) corporations is conducted under four foundational analytical assumptions:

1. **Balance-Sheet Integrity and Asset Reconciliation:** A reconciliation of reported balance-sheet asset line-items revealed minor accounting discrepancies (notably for Hindustan Unilever Ltd, where disclosed components aggregate to Rs. 99,738 Cr against reported total assets of Rs. 79,738 Cr). To maintain empirical integrity, all calculations strictly adopt the reported **Total Assets** figure as the definitive asset pool.
2. **Uniform Required Rate of Return ($K_e = 14.0\%$):** Derived from the Capital Asset Pricing Model (CAPM : $K_e = R_f + \beta \times \text{ERP}$) using a 10-year Indian sovereign bond yield ($R_f \approx 6.96\%$) and sector risk premium, a standardized discount rate of **14.0%** is adopted across all ten firms to ensure cross-sectional comparability.
3. **Sustainable Growth Ceiling ($g = 12.0\%$):** The Gordon Growth Model mathematically requires $K_e > g$. For enterprises where 5-year historical PAT compound annual growth exceeds 14% (HUL at 15.8%, Britannia at 14.2%) or reflects supernormal expansion (Varun Beverages), a conservative perpetual growth ceiling of **12.0%** is enforced. For all other firms, the 5-year historical PAT CAGR serves as the perpetual growth proxy.
4. **Standardized Dividend Proxy (Proxy DPS = 70% × FY26 EPS):** In the absence of historical dividend cash flow schedules in the source dataset, a dividend payout ratio of **70%** is applied to projected FY26 EPS, matching established FMCG capital distribution norms.

1.2 Tri-Partite Valuation and Consensus Decision Architecture

Figure 1 illustrates the methodological pipeline linking input data, the three independent valuation engines, and the consensus decision filter.

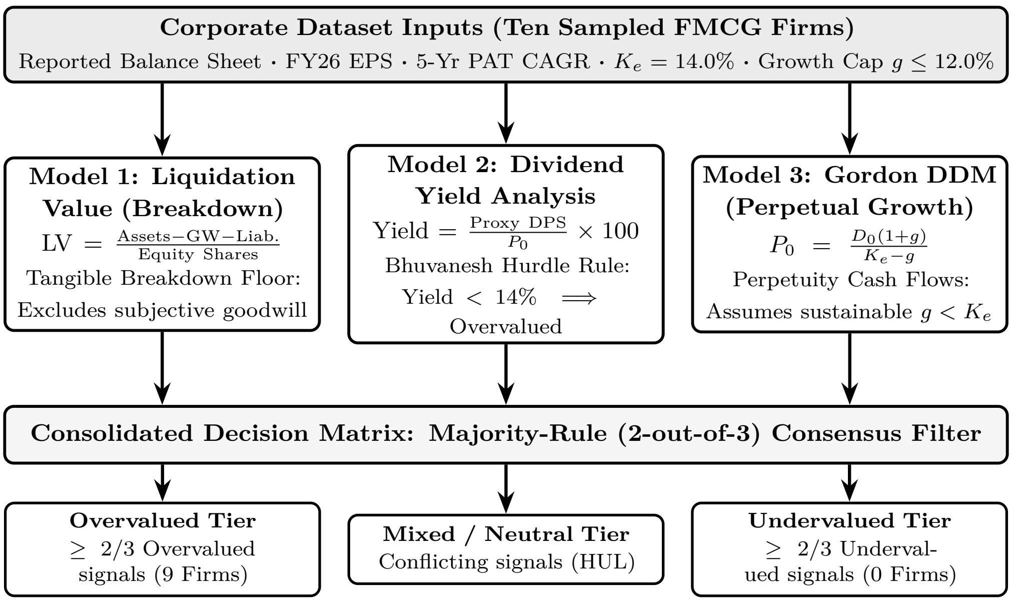

Figure 1: Tri-Partite Equity Valuation Architecture and Majority Consensus Pipeline.

2 Liquidation Value (Breakdown Value) Analysis

2.1 Theoretical Foundation and Rationale for Goodwill Exclusion

Under corporate valuation principles, the **Liquidation Value (Breakdown Value)** represents the estimated net cash surplus realized if an enterprise were to terminate operations immediately, liquidate all physical assets through orderly secondary transactions, and extinguish all senior liabilities.

Governing Equation: Liquidation Value per Share

$$\text{LV} = \frac{\text{Total Assets} - \text{Recorded Goodwill} - \text{Total Liabilities}}{\text{Equity Shares Outstanding}} \tag{1}$$

Goodwill is strictly deducted from the asset base because it represents capitalized historical acquisition premiums or intangible supernormal earning power. Under distress or bankruptcy conditions, commercial goodwill rapidly dissipates, possessing negligible liquidation recovery value.

2.2 Cross-Sectional Empirical Results

Table 1 summarizes the balance-sheet breakdown metrics, net realizable values, and resulting liquidation floors for the ten sampled FMCG corporations.

Company	**Total Assets** (Rs. Cr)	**Goodwill** (Rs. Cr)	**Total Liab.** (Rs. Cr)	**Net Realiz. Value** (Rs. Cr)	**Shares** (Cr)	**LV / Share** (Rs.)	**Market Price** (Rs.)
Hindustan Unilever	79,738	1,478	30,999	47,261	235	**Rs. 201**	Rs. 1,973
ITC Ltd	93,637	2,399	16,981	74,257	1,253	**Rs. 59**	Rs. 264
Nestle India	13,357	444	6,641	6,272	193	**Rs. 33**	Rs. 1,409
Varun Beverages	25,541	0	7,639	17,902	676	**Rs. 26**	Rs. 204
Britannia Industries	9,730	1,380	4,648	3,702	24	**Rs. 154**	Rs. 5,120
Marico Ltd	9,950	557	5,183	4,210	130	**Rs. 32**	Rs. 814
Tata Consumer	34,279	2,820	11,264	20,195	99	**Rs. 204**	Rs. 1,010
Godrej Consumer	13,365	2,948	5,573	4,844	102	**Rs. 47**	Rs. 881
Dabur India	17,480	1,287	6,238	9,955	177	**Rs. 56**	Rs. 382
Colgate-Palmolive	3,408	47	1,394	1,967	27	**Rs. 73**	Rs. 1,843

Table 1: Cross-Sectional Liquidation Value (Breakdown Value) Valuation Summary.

2.3 Analytical Interpretation and Valuation Multiples

1. **Universal Overvaluation Relative to Breakdown Floor:** All ten firms trade at substantial multiples above liquidation floors, ranging from 4.5× (ITC Ltd) to an extreme 42.7× (Nestle India Ltd).
2. **Capital Structure and Downside Protection: ITC Ltd** (P_0 = Rs. 264, LV = Rs. 59) and **Tata Consumer** (P_0 = Rs. 1,010, LV = Rs. 204, 4.9× multiple), backed by Rs. 74,257 Cr and Rs. 20,195 Cr in net tangible assets respectively, offer the strongest balance-sheet cushion against solvency shocks.
3. **Asset-Light Franchise Premium:** Formulators like **Nestle India** (42.7×) and **Britannia** (33.2×) command immense multiples because tangible plant and machinery generate only a fraction of cash flows; value is driven by uncapitalized brand equity, pricing power, and distribution moats.

3 Dividend Yield Analysis and the Bhuvanesh Hurdle Criterion

3.1 Theoretical Premise and the Bhuvanesh Decision Rule

The **Dividend Yield** method evaluates the direct annual cash distribution an investor receives relative to the secondary market price paid per share. Under the analytical framework in the course notes (*Bhuvanesh Rule*), dividend yield acts as an immediate income hurdle: if the current dividend yield is less than the investor's opportunity cost of capital ($K_e = 14.0\%$), the asset fails to compensate for capital risk on immediate cash flow alone and is classified as **Overvalued**.

Governing Equation and Bhuvanesh Hurdle Criterion

$$\text{Current Yield} = \frac{\text{Proxy DPS}}{P_0} \times 100 \tag{2}$$

$$\text{Decision Rule :} \quad \begin{cases} \text{Current Yield} < K_e\,(14.0\%) \implies \textbf{Overvalued} \\ \text{Current Yield} \geq K_e\,(14.0\%) \implies \textbf{Undervalued} \end{cases} \tag{3}$$

3.2 Cross-Sectional Dividend Yield Results

Table 2 details the projected FY26 EPS, standardized 70% proxy DPS, current market prices, computed dividend yields, and comparative verdicts against the 14.0% hurdle rate.

Company	FY26 EPS (Rs.)	Proxy DPS (70%, Rs.)	Market Price (Rs.)	Current Yield (%)	Hurdle (K_e) (14.0%)	Yield Verdict
Hindustan Unilever	64.08	44.86	1,973	2.27%	14.0%	**Overvalued**
ITC Ltd	16.77	11.74	264	4.45%	14.0%	**Overvalued**
Nestle India	18.37	12.86	1,409	0.91%	14.0%	**Overvalued**
Varun Beverages	4.53	3.17	204	1.55%	14.0%	**Overvalued**
Britannia Industries	105.71	74.00	5,120	1.44%	14.0%	**Overvalued**
Marico Ltd	13.95	9.77	814	1.20%	14.0%	**Overvalued**
Tata Consumer	15.63	10.94	1,010	1.08%	14.0%	**Overvalued**
Godrej Consumer	14.86	10.40	881	1.18%	14.0%	**Overvalued**
Dabur India	10.56	7.39	382	1.93%	14.0%	**Overvalued**
Colgate-Palmolive	49.07	34.35	1,843	1.86%	14.0%	**Overvalued**

Table 2: Cross-Sectional Dividend Yield Analysis and Bhuvanesh Hurdle Verification.

3.3 Analytical Takeaways from the Yield Criterion

1. **100% Failure of Current Income Hurdle:** Under the strict Bhuvanesh criterion, **all ten companies are classified as Overvalued**. None provide an immediate cash yield remotely approaching the 14.0% benchmark.
2. **Yield Dispersion Across the Universe: ITC Ltd** offers the highest dividend yield at 4.45%, followed by **HUL** at 2.27%. Conversely, **Nestle India** generates the lowest yield at 0.91%, returning under one rupee per hundred rupees invested.
3. **Economic Rationale:** The substantial gap between current yields ($0.9\% - 4.5\%$) and K_e (14.0%) demonstrates that secondary markets do not treat FMCG shares as quasi-fixed-income instruments; market valuations price in long-term earnings compounding and future dividend growth.

4 Dividend Discount Model (Gordon Growth Model) and Intrinsic Value

4.1 Theoretical Framework and Growth-Capping Methodology

The **Gordon Growth Model (DDM)** evaluates equity by discounting an infinite stream of future dividend distributions growing at a constant sustainable rate g:

$$P_0 = \sum_{t=1}^{\infty} \frac{D_t}{(1+K_e)^t} = \frac{D_1}{K_e - g} = \frac{D_0(1+g)}{K_e - g} \tag{4}$$

A key mathematical prerequisite is that the discount rate must strictly exceed the perpetual growth rate ($K_e > g$). If $g \geq K_e$, the valuation series diverges to infinity.

Governing Equation: Gordon Growth Model

$$P_0 = \frac{D_1}{K_e - g} = \frac{\text{Proxy DPS} \times (1+g)}{K_e - g} \tag{5}$$

Methodological Capping at $g = 12.0\%$: Historical 5-year PAT compound growth for **Hindustan Unilever** (15.8%), **Britannia Industries** (14.2%), and **Varun Beverages** exceed or approach the 14.0% hurdle. Because no enterprise can grow indefinitely faster than its cost of capital, an analytical ceiling of **12.0%** is applied to these three firms. For other firms, actual 5-year PAT CAGR is adopted.

4.2 Cross-Sectional DDM Intrinsic Valuation Results

Table 3 presents next-period expected dividends (D_1), assumed perpetual growth rates, computed DDM values, and resulting verdicts against current market prices.

Company	**Proxy DPS** (Rs.)	**Assumed g** (%)	**K_e** (%)	**D_1** (Rs.)	**DDM Value** (P_{DDM}, Rs.)	**Market Price** (Rs.)	**DDM Verdict**
Hindustan Unilever	44.86	12.00% (Cap)	14.0%	50.24	**Rs. 2,512**	Rs. 1,973	**Undervalued**
ITC Ltd	11.74	7.87%	14.0%	12.66	**Rs. 206**	Rs. 264	**Overvalued**
Nestle India	12.86	10.34%	14.0%	14.19	**Rs. 388**	Rs. 1,409	**Overvalued**
Varun Beverages	3.17	12.00% (Cap)	14.0%	3.55	**Rs. 178**	Rs. 204	**Overvalued**
Britannia Industries	74.00	12.00% (Cap)	14.0%	82.88	**Rs. 4,144**	Rs. 5,120	**Overvalued**
Marico Ltd	9.77	9.62%	14.0%	10.71	**Rs. 245**	Rs. 814	**Overvalued**
Tata Consumer	10.94	11.11%	14.0%	12.16	**Rs. 421**	Rs. 1,010	**Overvalued**
Godrej Consumer	10.40	0.62%	14.0%	10.46	**Rs. 78**	Rs. 881	**Overvalued**
Dabur India	7.39	1.77%	14.0%	7.52	**Rs. 61**	Rs. 382	**Overvalued**
Colgate-Palmolive	34.35	5.29%	14.0%	36.17	**Rs. 415**	Rs. 1,843	**Overvalued**

Table 3: Cross-Sectional Dividend Discount Model (DDM) Intrinsic Valuation.

4.3 Analytical Insights and Valuation Gaps

1. **HUL as the Sole Undervalued Firm:** HUL is the only company whose DDM intrinsic value (Rs. 2,512) exceeds market price (Rs. 1,973), showing a +27.3% **discount** under the capped 12% assumption.
2. **Nine Overvalued Enterprises:** The remaining nine firms are classified as **Overvalued**, led by **Godrej Consumer** (trading at 11.3× DDM value), **Dabur** (6.3×), **Colgate** (4.4×), and **Nestle** (3.6×).
3. **Spread Sensitivity ($K_e - g$):** As the spread ($K_e - g$) narrows to 2.0%, intrinsic values expand rapidly, illustrating the inherent model fragility and motivating sensitivity analysis.

5 Sensitivity Analysis: Hindustan Unilever Break-Even Dynamics

5.1 Algebraic Derivation of the Break-Even Perpetual Growth Rate

Because the Gordon Growth Model displays asymptotic convexity as $g \to K_e$, the valuation of Hindustan Unilever (Rs. 2,512 at $g = 12.0\%$ vs Market Price of Rs. 1,973) is subjected to rigorous sensitivity analysis. We analytically solve for the **break-even growth rate (g^*)** where intrinsic value exactly equates to the current market quotation:

$$P_0 = \frac{\text{Proxy DPS} \times (1+g)}{K_e - g} \implies 1,973 = \frac{44.86\,(1+g)}{0.14 - g} \tag{6}$$

Expanding and solving for g:

$$1,973 \times (0.14 - g) = 44.86\,(1+g) \implies 276.22 - 1,973\,g = 44.86 + 44.86\,g \tag{7}$$

$$231.36 = 2,017.86\,g \implies g^* = \frac{231.36}{2,017.86} \approx \mathbf{11.46\%} \tag{8}$$

HUL Growth-Rate Decision Boundaries

$g > 11.46\%$: Undervalued · **$g = 11.46\%$: Fairly Valued** · **$g < 11.46\%$: Overvalued**

5.2 Graphic Visualization of DDM Growth Sensitivity

Figure 2 plots HUL's DDM intrinsic value across a plausible perpetual growth domain (8.0% to 12.8%) against the current secondary market price of Rs. 1,973.

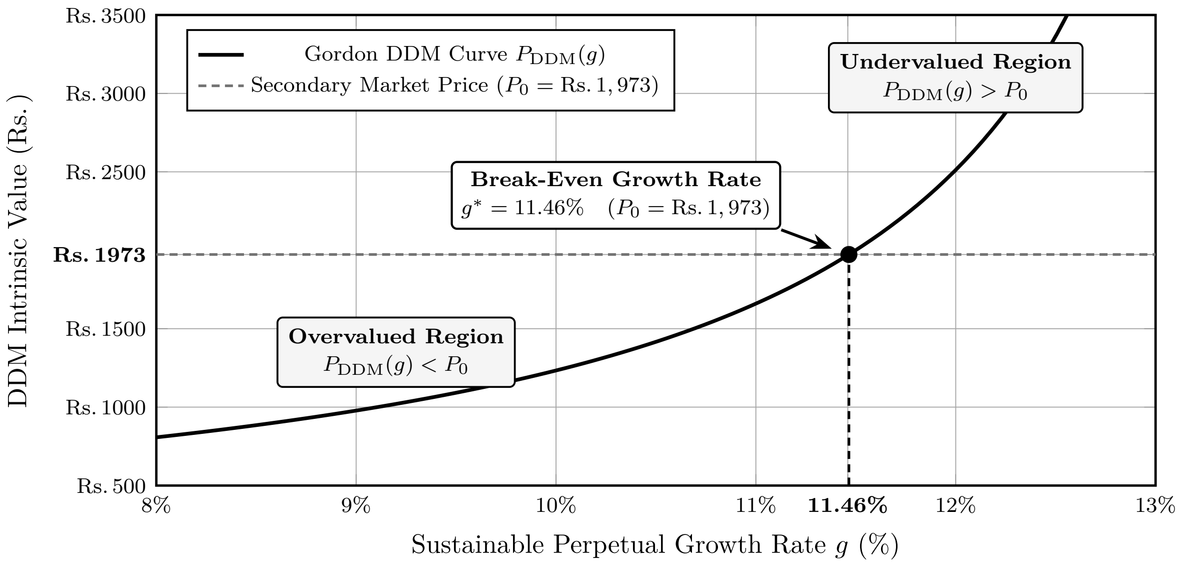

Figure 2: Hindustan Unilever: DDM Intrinsic Value Sensitivity to Growth Rate g ($K_e = 14.0\%$).

5.3 Empirical Implications of Growth Sensitivity

The empirical takeaway is that HUL's apparent undervaluation is exceptionally fragile: a minor deceleration of just **54 basis points** (from the assumed 12.00% to 11.46%) completely eliminates the margin of safety, rendering the corporation overvalued under the Gordon Growth framework.

6 Consolidated Decision Matrix and Majority Classification

6.1 Consensus Framework: Majority-Rule (2-out-of-3) Methodology

To synthesize the empirical findings across the three independent valuation methodologies, this study implements a formal **majority-rule (2-out-of-3) decision engine**:

- **Overvalued:** Assigned if ≥ 2 of the 3 valuation models indicate overvaluation.
- **Undervalued:** Assigned if ≥ 2 of the 3 valuation models indicate undervaluation.
- **Mixed / Neutral:** Assigned when signals conflict without a two-thirds majority.

Company	Liquidation Value Floor	Dividend Yield ($K_e = 14\%$ Hurdle)	Gordon DDM (Intrinsic Value)	Consensus (2/3) Classification
Hindustan Unilever	Overvalued	Overvalued	Undervalued	**Mixed / Neutral**
ITC Ltd	Overvalued	Overvalued	Overvalued	**Overvalued**
Nestle India	Overvalued	Overvalued	Overvalued	**Overvalued**
Varun Beverages	Overvalued	Overvalued	Overvalued	**Overvalued**
Britannia Industries	Overvalued	Overvalued	Overvalued	**Overvalued**
Marico Ltd	Overvalued	Overvalued	Overvalued	**Overvalued**
Tata Consumer	Overvalued	Overvalued	Overvalued	**Overvalued**
Godrej Consumer	Overvalued	Overvalued	Overvalued	**Overvalued**
Dabur India	Overvalued	Overvalued	Overvalued	**Overvalued**
Colgate-Palmolive	Overvalued	Overvalued	Overvalued	**Overvalued**

Table 4: Cross-Sectional Valuation Consensus Matrix under Study Assumptions.

7 Academic Conclusion and Master Formula Reference

7.1 Strategic Synthesis of Empirical Findings

1. **Liquidation Floor Disconnect:** All ten enterprises trade at substantial multiples (4.5× to 42.7×) above tangible liquidation worth. FMCG valuations reflect off-balance-sheet intangible moats (brand loyalty, negative working capital cycles, distribution) rather than physical asset backing.
2. **Dividend Yield Deficit:** With dividend yields spanning 0.91% to 4.45%, zero firms meet the 14.0% cost-of-equity hurdle, proving that market prices discount growth rather than current yield.
3. **Final Consensus Verdict:** By unanimous majority rule, **nine of the ten corporations are classified as Overvalued**. **Hindustan Unilever** presents a **Mixed / Neutral** profile; however, because its break-even growth buffer is a razor-thin 54 basis points (11.46%), any minor deceleration in consumer demand renders HUL overvalued as well.

7.2 Master Reference Table: Valuation Equations

Table 5 consolidates all governing mathematical formulations utilized throughout this study.

Valuation Metric	Mathematical Formulation	Operational Financial Purpose
Liquidation Value	$\text{LV} = \dfrac{\text{Total Assets} - \text{Goodwill} - \text{Liab.}}{\text{Shares Outstanding}}$	Demarcates the hard tangible breakdown floor.
Dividend Yield	$\text{Yield} = \dfrac{\text{Annual Proxy DPS}}{P_0} \times 100$	Evaluates direct cash return against 14% hurdle.
Gordon Growth Model	$P_0 = \dfrac{D_1}{K_e - g} = \dfrac{D_0(1+g)}{K_e - g}$	Values equity on perpetual growing dividend stream.
Expected Dividend (D_1)	$D_1 = D_0 \times (1+g)$	Forecasts immediate next-period cash distribution.
Cost of Equity (K_e)	$K_e = R_f + \beta \times [E(R_m) - R_f]$	Quantifies shareholder opportunity cost of capital.

Table 5: Consolidated Mathematical Formulations for Indian FMCG Equity Valuation.